HAPPIEST HEALTH DIGITAL TWIN CHALLENGE 2026

NoctTwin: Bedtime Digital Twin for Nocturnal Hypoglycemia Triage in Sulfonylurea-Treated T2DM

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Core Innovations:

- Zero-leakage bedtime machine learning risk prediction evaluated via patient-grouped 5-fold cross-validation.
- Blended Synchronised Forecaster (tau = 3.305h) beating persistence and historical profiles at 1h, 3h, and 6h horizons.
- Mechanistic two-compartment ODE evaluating what-if bedtime carbohydrate yields and secretagogue scaling.
- Held-out split-conformal uncertainty intervals (+/-26.3 mg/dL at 80% coverage, +/-37.6 mg/dL at 90% coverage).
- Transparent cold-start policy: automatic fallback to CGM-only Model B and population-average profile on first nights.

Dataset Realities, Regimen Audit & Artifact Filtration

Shanghai_T2DM Inpatient Dataset (Zhao et al. 2023):
• 100 hospitalized T2DM patients, 109 sessions, 1,181 valid bedtime nights.
• Hospital Inpatient Reality: Standardized hospital diet, reduced daytime ambulation, strictly scheduled sleep.
• Compression Artifact Filtration: FreeStyle Libre sensors produce artificial rapid nocturnal dips when slept on.
• Primary Clinical Target: >=2 consecutive readings <70 mg/dL between 23:00 and 06:00 (>=30 min duration).
• Prevalence: 124 nights (10.5%) across 45 patients; eliminated 24 single-point false compression dips.
• Severe Hypoglycemia (<54 mg/dL): 49 nights (4.1% of all monitored nights).

Canonical Regimen Breakdown & Event Concentration:
• SU without Insulin: 10 patients | 124 nights | 24 events (19.4%)
• SU with Insulin: 8 patients | 123 nights | 8 events (6.5%)
• Insulin without SU: 51 patients | 520 nights | 38 events (7.3%)
• Neither (Metformin/Other): 31 patients | 414 nights | 54 events (13.0%)
• Outlier Sensitivity: Top 10% of patients drive 48.4% of all lows. Patient 2066 had 9/13 nights in hypo (69.2%); excluding Pt 2066 drops SU rate to 13.5%. All SU findings are labeled exploratory.

Twin Architecture: Synchronised Forecast & Mechanistic Countermeasures

Blended Synchronised Forecaster (tau = 3.305 hours):

• *forecast(t) = profile(t) + (G_bedtime - profile(22:00)) * exp(-t / tau)*

• Solves Horizon Dilemma: Persistence wins at 1h; Historical Profile wins at 6h.

• Evaluated strictly on held-out patients across 5 GroupKFold splits:

- 1h (23:00): MAE 14.83 mg/dL vs Persistence 18.48 vs Profile 24.62 (Delta = -3.64 vs Persistence)
- 3h (01:00): MAE 19.35 mg/dL vs Profile 21.69 vs Persistence 31.97 (Delta = -2.33 vs Profile)
- 6h (04:00): MAE 16.05 mg/dL vs Profile 17.18 vs Persistence 36.98 (Delta = -1.13 vs Profile)

All paired difference 95% bootstrap CIs strictly exclude 0.

Two-Compartment Mechanistic ODE Countermeasures:

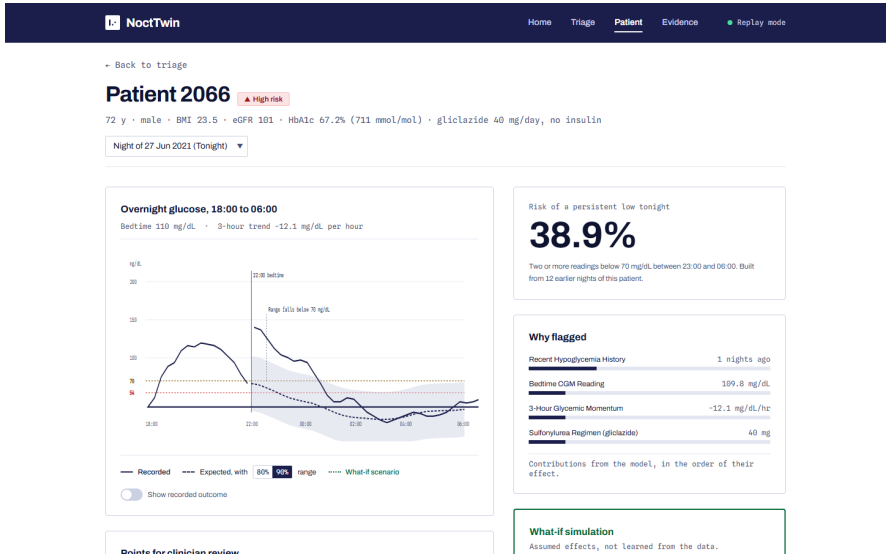
• Gut absorption (k_abs=0.035/min, S_snack=4.2 mg/dL/g) and secretagogue decay (tau_SU=4.0h) compute countermeasure deltas overlaid onto synchronised baseline.

Split-Conformal Prediction Coverage (200 Patient-Level Splits):

• 80% Target: Mean coverage 79.93% | Margin +/-26.3 mg/dL

• 90% Target: Mean coverage 89.84% | Margin +/-37.6 mg/dL

• Conditional on Forecast < 110 mg/dL: 95.2% coverage.



Model Validation: What Helped and What Did Not

Primary Benchmark: Nights 1+ (20 Repeated GroupKFold Runs, N=1,081 nights, 101 events):
• Group G (CGM + History): AUROC 0.832 (range 0.824-0.840), AUPRC 0.476 (range 0.417-0.496), Brier 0.065 (range 0.064-0.068).
• Group B (CGM-Only): AUROC 0.736 (range 0.707-0.753), AUPRC 0.354 (range 0.252-0.392), Brier 0.073 (range 0.071-0.078).
• Paired Advantage: Delta AUROC = +0.097 [+0.051, +0.149], Delta AUPRC = +0.123 [+0.047, +0.191], Delta Brier = -0.008 [-0.014, -0.004] (strictly positive gains; 95% CIs exclude 0).
• Secondary Full Cohort (N=1,181): Group G AUROC 0.838 vs Group B 0.772 (Delta AUROC = +0.066 [+0.033, +0.105]).

What Helped Most:
• Days since last nocturnal low and personal median historical nadir.
• Bedtime CGM momentum (3h linear slope and bedtime glucose level).

What Did Not Help (Static EHR Labs):
• Adding 28 routine admission labs (Group E) collapsed linear model precision (Delta AUPRC = -0.108 [-0.159, -0.025]).
• Fingerprint Memorization: Static labs identify patients rather than dynamic physiology in small cohorts.

Cold-Start First Nights (Underpowered Sensitivity Analysis):
• On Night 1 (N=100 nights, 10 events), adding meds/labs gave no measurable benefit over acute CGM dynamics alone (Group B AUROC 0.813 vs C 0.783 vs D 0.747); analysis is underpowered with only 10 events.

False-Alarm Burden & Operational Clinical Operating Points

The Bayes 75% PPV Trap:

- At 10.5% cohort prevalence, achieving a 75% PPV target mathematically requires a positive likelihood ratio LR+ >= 25.57.
- Enforcing 75% PPV (actual 78.6%) forces the threshold to 0.475, causing sensitivity to collapse to 17.7% (missing 82.3% of true nocturnal lows!).
- A clinical tool that misses 82% of nocturnal lows is unsafe for hospital triage.

Operational Recommendation (80% Specificity Operating Point):

- Calibrated Bedtime Risk Threshold: 0.12
- Sensitivity: 75.0% (intercepts 3 out of every 4 nocturnal lows).
- Specificity: 80.0%.
- Positive Predictive Value (PPV): 30.6%.
- Number-Needed-to-Warn (NNW): 3.27 (one true low per ~3 flagged patients).
- Alert Burden: 1.8 alerts per patient-week.
- Clinical Safety Net: Avoids alarm fatigue while providing high nocturnal protection.

NOCTTWIN | CLINICIAN INTERFACE

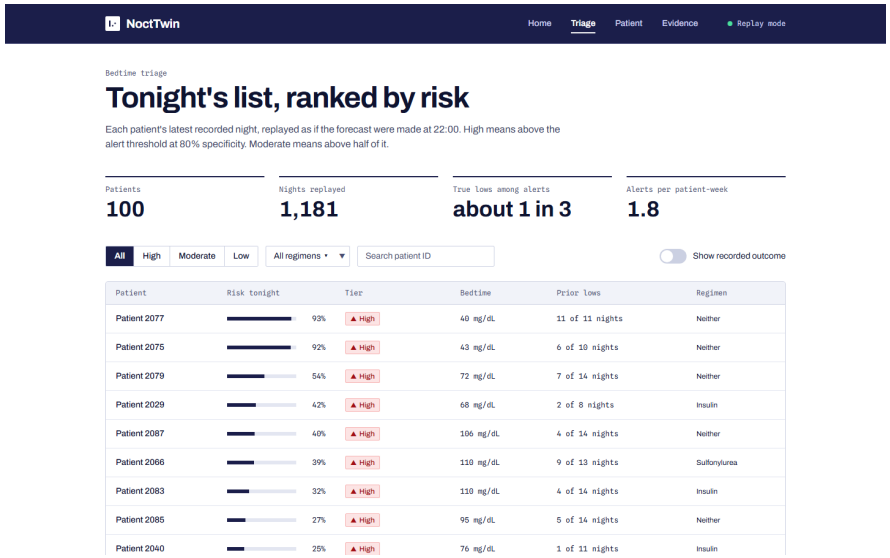
Product Implementation & Clinical Decision Flow

Production Architecture:
• FastAPI backend + React 18, Vite, TypeScript, Tailwind, Archivo & Spline Sans Mono fonts.

Screen 1: Bedtime Triage Board:
• Monotonic risk tiering (High >=0.126, Moderate >=0.063, Low <0.063).
• Blind prospective triage mode toggle hides recorded outcomes.
• Clean tabular layout with prior low counts and regimen chips.

Screen 2: Patient Twin View:
• 24h pre-anchor CGM + Blended Synchronised Forecaster + Conformal band.
• Cold-Start Fallback: Automatic fallback on first nights.
• Interactive What-If Simulation: Sliders for bedtime snack carbs and SU dose.
• Non-prescriptive deterministic points for clinician review.

Screen 3: Scientific Evidence Card:
• Hairline tables loaded from generated reports without hardcoding.



Assumptions, Limitations & Open Scientific Questions

Key Limitations Transparently Logged:

1. **Static Regimen Classification:** Medications are recorded per-patient at admission, not dynamically updated per-night.
2. **Confounding of History & Compression Posture:** History features cannot separate true recurrent hypoglycemia from habitual sleep posture (sleeping on sensor arm).
3. **Unresolved Night 0 Anomaly (Open Question):** Night 0 event prevalence is 2.46x higher than subsequent nights (23.0% vs 9.34%). Whether this represents FreeStyle Libre insertion trauma/first-day sensor lag or acute hospital admission stress/medication changes remains an unresolved open question.
4. **Inpatient Generalizability:** Shanghai_T2DM comprises hospitalized inpatients under standard hospital diet. Generalizability to free-living outpatients with variable evening meals and alcohol intake requires external ambulatory validation.
5. **Outlier Sensitivity in Small SU Subgroup:** 18 total SU patients; 10 SU without insulin. Results are exploratory and sensitive to single outlier patients (e.g., Pt 2066).
6. **Non-Validated What-If Countermeasures:** Physiological simulations represent hypotheses, not clinically approved intervention plans.

Roadmap: Path to Ambulatory Clinical Translation

Translation Milestones from Academic Prototype to Bedside CDS:

1. **Prospective Ambulatory Validation:** Multi-center validation on free-living outpatients wearing factory-calibrated sensors (Dexcom G7 / Abbott Libre 3) with wearable accelerometers to separate true hypoglycemia from compression artifacts.

2. **Randomized Bedtime Countermeasure Trial:** Clinical trial comparing NoctTwin bedtime snack guidance against standard care, tracking nocturnal lows and morning rebound hyperglycemia.

3. **Concurrent Nocturnal ECG Co-Monitoring:** Ingest continuous Holter / patch ECG telemetry to empirically evaluate QTc prolongation (>450ms) during deep nocturnal glucose troughs (<54 mg/dL).

4. **Native EHR / FHIR Integration:** Seamless integration into hospital electronic health records to ingest real-time bedside meal logs and bedtime capillary glucose verifications.

5. **Continuous Digital Twin Parameter Learning:** Bayesian update of patient absorption (k_abs) and secretagogue sensitivity (E_SU) over sequential nights as personal data accumulates.